RAPHAËL REYNOUARD

CONTACT INFO

E-mail raphael.reynouard@eawag.ch

Website https://rapfff.github.io

Address Office: FC-D09, Überlandstrasse 133, 8600 Dübendorf

Phone +41 58 765 6406

ABOUT ME

I am a post-doctoral researcher at eawag, the Swiss Federal Institute of Aquatic Science and Technology. I develop, design and implement machine learning methods applied on hydrological systems.
I completed my PhD at Reykjavik University (Iceland), focusing on machine learning techniques for Markov model learning, under the supervision of Anna Ingolfsdottir and Giovanni Bacci. My research interests are machine learning, hydrology, model checking, markov models, statistical modeling.

EDUCATION

PhD in Computer Science **2020-2023**
Háskólinn í Reykjavík | *Reykjavík, Iceland*

- Field: Machine Learning & Model Checking
- Thesis topic: "Machine Learning from a Model Checking Perspective"
- Supervisor: Anna Ingólfsdóttir.
- PhD Thesis: On learning stochastic models: from theory to practice

BA & MSc in Computer Science **2015-2020**
Université Libre de Bruxelles | *Brussels, Belgium*

- Graduated with *Disctinction*
- Thesis topic: "Learning of Non-Markovian Reward Functions"
- Supervisor: Jean-François Raskin.

PUBLICATIONS

R. Reynouard, A. Ingólfsdóttir, and G. Bacci (2023). "Jajapy: a learning library for stochastic models". In: *Quantitative Evaluation of Systems - 20th International Conference, QEST 2023, Antwerp, Belgium, September 18-23, 2023, Proceedings*. Ed. by Nils Jansen Mirco Tribastone. Lecture Notes in Computer Science. Springer.

Bacci, G., A. Ingólfsdóttir, K. G. Larsen, and **R. Reynouard** (2021). "Active Learning of Markov Decision Processes using Baum-Welch algorithm". In: *20th IEEE International Conference on Machine Learning and Applications, ICMLA 2021, Pasadena, CA, USA, December 13-16, 2021*. IEEE, pp. 1203–1208. DOI: 10.1109/ICMLA52953.2021.00195.

– (2023). "An MM Algorithm to Estimate Parameters in Continuous-time Markov Chains". In: *Quantitative Evaluation of Systems - 20th International Conference, QEST 2023, Antwerp, Belgium, September 18-23, 2023, Proceedings*. Ed. by Nils Jansen Mirco Tribastone. Lecture Notes in Computer Science. Springer.

Rens, G., J.-F. Raskin, **R. Reynouard**, and G. Marra (2021). "Online Learning of non-Markovian Reward Models". In: *Proceedings of the 13th International Conference on Agents and Artificial Intelligence, ICAART 2021, Volume 2, Online Streaming, February 4-6, 2021*. SCITEPRESS, pp. 74–86. DOI: 10.5220/0010212000740086.

TOOLS

Jajapy — **since 2021**

A Python learning library for Markov models

- Open source Python library
- Implementing several learning methods for stochastic models
- Compatible with the model checkers Storm and Prism
- Github: https://github.com/Rapfff/jajapy
- Documentation: https://jajapy.readthedocs.io/en/latest/index.html.

AWARDS

Babbage Prize — **2020**

- Each year, the Babbage Association awards the Babbage Prize to a student who has graduated from a Master's degree in Computer Science at ULB and who has distinguished himself as much by his commitment to the student community as by the excellence of his academic career.

MISCELLANEOUS

Mathematics tutoring — **since 2016**

- Tutoring in mathematics for middle and high school students
- More than 20 students.

Scout Leader — **2016-2020**

Belgium

- Scoutmaster of a Belgian group
- From 2018 to 2020, head of the boys' group (60-70 children).

SKILLS

●●● Python, LaTeX

●●○ Linux, R, PyTorch

● ○ ○ Arduino, Java

Languages

- French (Native)
- English (Full professional proficiency)
- Icelandic (Basic)

REFERENCES

Kim G. Larsen
Professor
Department of Computer Science
Aalborg University
kgl@cs.aau.dk
http://www.cs.aau.dk/~kgl

Giovanni Bacci
Associate Professor
Department of Computer Science
Aalborg University
giovbacci@cs.aau.dk
https://homes.cs.aau.dk/~giovbacci/

Anna Ingólfsdóttir
Professor
Department of Computer Science
Reykjavik University
anna.ingolfsdottir@gmail.com
http://kennarar.ru.is/annai/

Luca Aceto
Professor
Department of Computer Science
Reykjavik University
luca@ru.is
http://www.ru.is/faculty/luca/

Joost-Pieter Katoen
Professor
Department of Computer Science
RWTH Aachen University
katoen@cs.rwth-aachen.de
http://www-i2.informatik.rwth-aachen.de/~katoen/